

SP-13: Trace-norm-small perturbations preserve Hermitian spectral distributions

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This manuscript proves the complete Barbarino-Serra-Capizzano trace-norm perturbation conjecture. It was developed with substantial ChatGPT/Codex assistance at the author's request. A separate [independent Codex-agent mathematical review](#) returned PASS for the complete target; a [coordinating-agent audit](#) also found no gap. No external human peer review, formal proof-assistant verification, or priority determination is asserted. The [submission record](#) preserves the frozen source, original target and verification evidence.

1. Exact target

For $A_n \in \mathbb{C}^{n \times n}$ and measurable $f : [0, 1] \rightarrow \mathbb{R}$, the notation $\{A_n\} \sim_\lambda f$ means

$$\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{j=1}^n F(\lambda_j(A_n)) = \int_0^1 F(f(t)) dt \quad (F \in C_c(\mathbb{C})).$$

The functions are continuous, complex-valued, and compactly supported. Eigenvalues have algebraic multiplicity. The trace norm is $\|X\|_* = \sum_j \sigma_j(X)$; X^* is the conjugate transpose.

Theorem. Suppose $H_n = H_n^*$, $\{H_n\} \sim_\lambda f$, and $E_n \in \mathbb{C}^{n \times n}$ satisfies $\|E_n\|_*/n \rightarrow 0$. Then

$$\{H_n + E_n\} \sim_\lambda f.$$

There is no spectral-norm boundedness assumption on either sequence and no normality assumption on $H_n + E_n$. This is the complete canonical SP-13 target and Conjecture 1 in Section 6, printed page 29, of Barbarino-Serra-Capizzano [BSC20].

2. The external weak-type theorem

For a finite matrix X and $s > 0$, set

$$n_X(s) = \#\{j : \sigma_j(X) > s\}, \quad \|X\|_{1,\infty} = \sup_{s>0} s n_X(s).$$

Let $\mathcal{T}_+(X)$ retain the strictly upper-triangular entries of X and replace all other entries by zero.

Published weak-type estimate. There is an absolute constant C such that, for every matrix size and every complex matrix X ,

$$n_{\mathcal{T}_+(X)}(s) \leq \frac{C\|X\|_*}{s} \quad (s > 0). \quad (1)$$

This is the finite-matrix specialization of Randrianantoanina [R02, Theorem 4.8, printed page 23]. The source defines the weak- L^1 quasi-norm on printed page 12 by the distribution function used above. Its theorem applies to an arbitrary finite family of mutually orthogonal projections and has an **absolute** constant. Take the algebra $M_n(\mathbb{C})$ with its ordinary, unnormalized matrix trace, and its rank-one coordinate projections. The source uses strictly lower-triangular truncation; reversing the order of the projections gives $\mathcal{T}_+$. Thus (1) is uniform in n and imposes neither positivity nor self-adjointness on X .

Theorem 4.8 is explicitly attributed there to Dodds-Dodds-de Pagter-Sukochev [DDPS99, Theorem 1.4], and [R02] also supplies a proof from its noncommutative Hilbert-transform

estimate. We use this published result; we do not claim it as a new theorem or replace it by the weaker logarithmic trace-norm bound.

3. Two elementary spectral facts

For arbitrary matrices X, Y and $a, b > 0$,

$$n_{X+Y}(a+b) \leq n_X(a) + n_Y(b). \quad (2)$$

Indeed, remove the singular components of X larger than a , and of Y larger than b . The remaining sum has spectral norm at most $a+b$, while the removed sum has rank at most the right side. The min-max characterization of singular values proves (2). Also

$$n_X(a) \leq \frac{\|X\|_*}{a}. \quad (3)$$

Lemma. Let A_n, B_n be Hermitian matrices of size n . If

$$\frac{n_{A_n-B_n}(s)}{n} \rightarrow 0 \quad \text{for every fixed } s > 0, \quad (4)$$

then for every $g \in C_c(\mathbb{R})$,

$$\frac{1}{n} \operatorname{tr} g(A_n) - \frac{1}{n} \operatorname{tr} g(B_n) \rightarrow 0. \quad (5)$$

No bound on $\|A_n\|$ or $\|B_n\|$ is required.

Proof. Fix $s > 0$. Split the Hermitian matrix $A_n - B_n$ by its spectral decomposition into $R_n + S_n$, where R_n contains the eigenvalues with absolute value greater than s . Then

$$\operatorname{rank} R_n = n_{A_n-B_n}(s) = o(n), \quad \|S_n\| \leq s.$$

For Hermitian matrices differing by rank at most r , their eigenvalue counting functions differ by at most r at every real threshold. This follows from the min-max principle by restricting a trial subspace to the kernel of the rank- r difference. Consequently, for $g \in C_c^1(\mathbb{R})$, integration by parts against the finite eigenvalue counting measures gives

$$|\operatorname{tr} g(A_n) - \operatorname{tr} g(B_n + S_n)| \leq \operatorname{rank} R_n \int_{\mathbb{R}} |g'(x)| dx. \quad (6)$$

The eigenvalues of $B_n + S_n$ and B_n , arranged in increasing order, differ by at most s , again by min-max. With

$$\omega_g(s) = \sup_{|x-y| \leq s} |g(x) - g(y)|,$$

their normalized trace difference is at most $\omega_g(s)$. Thus the limsup of the absolute value in (5) is at most $\omega_g(s)$. Compactly supported continuous functions are uniformly continuous, so letting $s \downarrow 0$ proves (5) for $g \in C_c^1(\mathbb{R})$. Every $g \in C_c(\mathbb{R})$ can be uniformly approximated by such functions; each normalized trace changes by at most the uniform error. This proves the lemma, also for complex-valued g . $\square$

4. Schur decomposition and the finite-dimensional estimates

Fix n , write $H = H_n$, $E = E_n$, and let $q = \|E\|_*$. Take any unitary Schur decomposition

$$U^*(H + E)U = T = D + iJ + N,$$

where D and J are real diagonal matrices and N is strictly upper triangular. The diagonal entries of $D + iJ$ are exactly the eigenvalues of $H + E$, counted with algebraic multiplicity. Set

$$G = U^* H U, \quad Z = U^* E U, \quad K = \operatorname{Im} T = \frac{T - T^*}{2i}.$$

Because G is Hermitian,

$$K = \operatorname{Im} Z, \quad \|K\|_* \leq \|Z\|_* = q. \quad (7)$$

For a strictly upper-triangular entry, $K_{ij} = N_{ij}/(2i)$. Therefore

$$N = 2i \mathcal{T}_+(K).$$

The dimension-independent estimate (1) yields

$$n_N(s) \leq \frac{2Cq}{s}. \quad (8)$$

Applying (2) to $(N + N^*)/2$ gives

$$n_{\operatorname{Re} N}(s) \leq 2n_N(s) \leq \frac{4Cq}{s}. \quad (9)$$

Taking Hermitian parts in $G + Z = D + iJ + N$ gives

$$D - G = \operatorname{Re} Z - \operatorname{Re} N.$$

Since $\|\operatorname{Re} Z\|_* \leq q$, equations (2), (3), and (9) show, for every $s > 0$,

$$n_{D-G}(s) \leq n_{\operatorname{Re} Z}(s/2) + n_{\operatorname{Re} N}(s/2) \leq \frac{(2 + 8C)q}{s}. \quad (10)$$

Finally, the diagonal of K is J , and

$$\sum_{j=1}^n |J_{jj}| \leq \|K\|_* \leq q. \quad (11)$$

For clarity, the first inequality needs no bound on a triangular projection: if S is the diagonal matrix with entries $\operatorname{sgn}(K_{jj})$, then $\|S\| \leq 1$ and $\sum_j |K_{jj}| = \operatorname{tr}(SK) \leq \|K\|_*$. Equivalently it is trace-norm contractivity of diagonal pinching.

All estimates (7)–(11) hold for every finite n , independently of H , the Schur ordering, diagonalizability, or the spectral norm of E .

5. Passage to the prescribed test functions

Return to the sequences, so $q_n/n \rightarrow 0$. By (10), for every fixed $s > 0$,

$$\frac{n_{D_n - U_n^* H_n U_n}(s)}{n} \rightarrow 0.$$

The lemma implies that D_n and H_n have asymptotically equal normalized traces for all $g \in C_c(\mathbb{R})$. In particular, for a prescribed $F \in C_c(\mathbb{C})$, the restriction $g(x) = F(x)$ to the real axis belongs to $C_c(\mathbb{R})$, and

$$\frac{1}{n} \sum_j F((D_n)_{jj}) - \frac{1}{n} \sum_j F(\lambda_j(H_n)) \rightarrow 0. \quad (12)$$

Write $d_{nj} = (D_n)_{jj}$ and $y_{nj} = (J_n)_{jj}$. Equation (11) implies

$$\#\{j : |y_{nj}| > s\} \leq \frac{q_n}{s}.$$

Every continuous compactly supported function on $\mathbb{C}$ is uniformly continuous on the whole plane. Defining its modulus $\omega_F(s)$ using Euclidean distance in $\mathbb{C}$, we obtain

$$\begin{aligned} & \left| \frac{1}{n} \sum_j F(d_{nj} + iy_{nj}) - \frac{1}{n} \sum_j F(d_{nj}) \right| \\ & \leq \omega_F(s) + 2\|F\|_\infty \frac{q_n}{ns}. \end{aligned} \quad (13)$$

For every fixed $s > 0$ the second term tends to zero. Then $s \downarrow 0$ makes the first term tend to zero. The values $d_{nj} + iy_{nj}$ are the eigenvalues of $H_n + E_n$ with their algebraic multiplicities. Combining (12), (13), and the assumed spectral distribution of H_n proves

$$\lim_{n \rightarrow \infty} \frac{1}{n} \sum_j F(\lambda_j(H_n + E_n)) = \int_0^1 F(f(t)) dt.$$

This holds for every $F \in C_c(\mathbb{C})$ and is exactly the asserted conclusion. $\square$

6. Scope and attribution

The proof allows arbitrary complex perturbations and arbitrary real measurable symbols on the canonical domain. Its only smallness assumption is $q_n/n \rightarrow 0$. Large exceptional singular values and eigenvalues are handled by rank bounds and compactly supported test functions, so no uniform spectral bound enters implicitly.

The published weak-type theorem retains its attribution to Randrianantoanina and Dodds, Dodds, de Pagter and Sukochev. Barbarino and Serra-Capizzano retain credit for the conjecture and preceding perturbation results. This proof applies the dimension-independent triangular-truncation estimate to the imaginary Hermitian part of a Schur form.

References

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